# Semantic-Guided Autonomous Mapping: A Survey of Vision-Language-Action Models and Their Integration with Visual SLAM for Predictive Next-Best-View Exploration on Edge Devices

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***Abstract*—Autonomous exploration of complex infrastructures demands the seamless integration of high-fidelity spatial perception with cognitive reasoning capabilities. This survey systematically reviews the convergence of three rapidly evolving research threads: (i) visual Simultaneous Localisation and Mapping (ViSLAM) augmented with 3D Gaussian Splatting (3DGS) for real-time, photorealistic scene reconstruction; (ii) Vision-Language-Action (VLA) foundation models that endow robots with grounded spatial understanding and natural-language instruction following; and (iii) predictive Next-Best-View (Pred-NBV) planners that minimise exploration entropy under energy-constrained budgets. We articulate how VLA-driven semantic intelligence can proactively guide metric reconstruction through closed-loop Pred-NBV strategies, reducing flight effort while maximising surface coverage. Moreover, we analyse the feasibility of deploying these architectures on edge-computing hardware (e.g., NVIDIA Jetson) via parameter-efficient fine-tuning techniques such as Low-Rank Adaptation (LoRA) and 8-bit quantisation. A comparative analysis of reconstruction quality, exploration efficiency, cognitive performance, and hardware utilisation metrics consolidates the state of the art and identifies open challenges toward fully autonomous, semantically aware mobile robots and unmanned aerial vehicles (UAVs).**

***Index Terms*—survey, visual SLAM, 3D Gaussian Splatting, vision-language-action models, next-best-view planning, edge computing, autonomous exploration, UAV**

## I. Introduction

Autonomous inspection of built environments—ranging from industrial plants and bridges to disaster-stricken buildings—requires robotic platforms capable of constructing dense, geometrically faithful maps while reasoning about *what* to observe next and *why*. Traditional pipelines decouple spatial perception from high-level cognition: a Simultaneous Localisation and Mapping (SLAM) front-end densifies the map, and a separate planner selects viewpoints purely on the basis of geometric information gain [1]. This decoupling introduces two critical limitations. First, the planner remains "blind" to semantic context—it cannot distinguish a crack in a load-bearing column from an irrelevant texture on a floor tile. Second, the computational cost of both dense reconstruction and large vision-language models (VLMs) has historically confined these systems to cloud-based deployments, precluding real-time autonomy on weight- and power-constrained platforms such as unmanned aerial vehicles (UAVs).

Recent advances across three complementary fronts motivate a re-examination of this architecture. On the reconstruction side, 3D Gaussian Splatting (3DGS) has emerged as a compelling alternative to Neural Radiance Fields (NeRF), achieving superior photorealistic rendering at frame rates exceeding 100 fps while maintaining competitive geometric accuracy [2], [3]. On the cognitive side, Vision-Language-Action (VLA) models—exemplified by architectures grounded in Qwen-VL [4] and embodied reasoning frameworks such as Lumo-1 [5]—have demonstrated the ability to interpret complex natural-language instructions with spatial awareness, achieving success rates above 90% in task-switching benchmarks [6]. On the planning side, predictive Next-Best-View (Pred-NBV) and generalizable NBV (GenNBV) algorithms leverage learned priors to observe up to 25.46% more surface points within the same time budget compared with classical information-gain planners [1], [7].

Despite these individual breakthroughs, no existing survey comprehensively addresses their *joint* integration—specifically, how VLA-driven semantic reasoning can close the loop with metric reconstruction and predictive exploration under edge-computing constraints. This paper fills that gap by:

1) Providing a structured taxonomy of modern VLA models and their coupling with real-time 3DGS-based SLAM systems (e.g., MCGS-SLAM [8], MVS-GS [9]).
2) Analysing the computational feasibility of deploying these architectures on embedded hardware through Low-Rank Adaptation (LoRA) and quantisation strategies [10].
3) Proposing a reference closed-loop architecture that unifies dense metric perception (3DGS), semantic cognition (VLA), and active exploration (Pred-NBV / GenNBV) for energy-efficient autonomous mapping.

The remainder of this paper is organised as follows. Section II surveys the foundational technologies. Section III presents the proposed semantic-guided mapping framework. Section IV provides a quantitative comparative analysis. Section V discusses open challenges, and Section VI concludes the paper.

## II. Background and Related Work

### A. Visual SLAM and 3D Reconstruction

Visual Simultaneous Localisation and Mapping (ViSLAM) constitutes the perceptual backbone of autonomous mobile robots. Classical feature-based methods (e.g., ORB-SLAM) track sparse landmarks to estimate camera trajectories, whereas direct methods minimise photometric error over dense or semi-dense image regions. More recently, learning-based SLAM systems have incorporated deep feature extraction and end-to-end pose regression, improving robustness under challenging illumination and texture-poor conditions.

A critical evolution is the integration of explicit 3D reconstruction pipelines that produce dense, renderable scene representations. Multi-view stereo (MVS) approaches, such as MVS-GS [9], combine online depth estimation with Gaussian-based scene representations to achieve high-quality mapping at interactive rates. Multi-camera SLAM frameworks such as MCGS-SLAM [8] further extend fidelity by fusing observations from multiple viewpoints into a unified Gaussian map, enabling high-fidelity reconstruction suitable for downstream inspection tasks.

Key evaluation metrics in this domain include the Absolute Trajectory Error (ATE) for localisation accuracy, the Chamfer Distance (CD) for geometric fidelity, and rendering quality indicators such as PSNR, SSIM, and LPIPS [3].

### B. 3D Gaussian Splatting

3D Gaussian Splatting (3DGS) represents scenes as collections of anisotropic 3D Gaussians, each parameterised by a mean position, covariance matrix, opacity, and view-dependent colour encoded via spherical harmonics. Rendering is performed through differentiable $\alpha$-blending of the splatted Gaussians, enabling real-time novel-view synthesis at rates exceeding 100 fps—and up to 200 fps in optimised configurations—while maintaining photorealistic quality [11].

Compared with NeRF-based implicit representations, 3DGS offers several advantages for robotic applications: (i) explicit geometry that can be directly queried for collision checking and path planning; (ii) incremental scene updates without full network retraining; and (iii) significantly lower inference latency. In urban-scale scenes, 3DGS achieves PSNR values of up to 28.93 dB and SSIM scores of 0.918, with LPIPS as low as 0.056, outperforming NeRF baselines across all perceptual metrics [3], [12].

Recent works have extended 3DGS to large-scale aerial-to-ground reconstruction. Horizon-GS [12] unifies multi-altitude observations into a single Gaussian field, enabling seamless transitions between bird's-eye and street-level views. Feed-forward variants such as DrivingForward [11] eliminate per-scene optimisation entirely, achieving reconstruction latencies between 0.29 and 0.63 seconds per scene—a critical enabler for real-time autonomous exploration.

A comprehensive survey by Liu et al. [2] traces the evolution from classical multi-view geometry through NeRF to 3DGS, consolidating the theoretical underpinnings and benchmarking methodologies that inform the present work.

### C. Vision-Language-Action Models

Vision-Language-Action (VLA) models extend large vision-language models (VLMs) by incorporating an action output head, enabling end-to-end mapping from visual observations and language instructions to robot control commands. These models serve as the "cognitive brain" of the robotic system, interpreting high-level mission objectives (e.g., "inspect the underside of the bridge deck for corrosion") and translating them into actionable behaviours.

State-of-the-art VLA architectures build upon powerful VLM backbones. Qwen2.5-VL [4] demonstrates spatial reasoning accuracies between 75.6% and 86.3% on the EmbSpatial benchmark, providing a robust foundation for grounded instruction following. Lumo-1 [5], developed for embodied reasoning, integrates purposeful robotic control with multi-modal perception, enabling complex manipulation and navigation tasks through chain-of-thought reasoning.

A key challenge for deployment on mobile platforms is task switching. SwitchVLA [6] addresses this by introducing execution-aware mechanisms that achieve success rates above 90% when transitioning between heterogeneous tasks—a capability essential for multi-objective inspection missions where the robot must dynamically reprioritise between mapping, anomaly inspection, and return-to-base objectives.

### D. Next-Best-View Planning

Next-Best-View (NBV) planning determines the optimal sensor placement to maximise information gain per observation, subject to motion and energy constraints. Classical NBV approaches rely on volumetric information-gain heuristics computed over an occupancy grid, selecting the viewpoint that maximises expected entropy reduction.

Pred-NBV [1] advances this paradigm by learning to *predict* the information gain of candidate viewpoints using a neural network trained on partial reconstructions. This prediction-guided strategy observes up to 25.46% more surface points than conventional planners within the same temporal budget, while simultaneously reducing unnecessary exploratory motions.

GenNBV [7] further generalises NBV planning across object categories by training a policy network on diverse 3D assets, enabling zero-shot transfer to unseen objects. The resulting policy optimises the trade-off between flight path length and surface coverage, a metric of paramount importance for battery-constrained UAVs.

Zhang et al. [10] demonstrate the integration of NBV-driven exploration with 3DGS-based reconstruction for semantic indoor mapping using autonomous UAVs, representing one of the first end-to-end systems that couples active view planning with Gaussian scene representations.

## III. Semantic-Guided Autonomous Mapping

### A. Closed-Loop Architecture

The central contribution of this survey is the articulation of a closed-loop architecture that unifies three subsystems: (i) a dense metric perception module based on ViSLAM with 3DGS scene representation, (ii) a cognitive reasoning module instantiated by a VLA model, and (iii) a predictive exploration module driven by Pred-NBV or GenNBV algorithms.

The perception module continuously maintains an incremental 3DGS map, providing the VLA module with rendered views, depth estimates, and a semantic uncertainty map. The VLA module interprets mission-level objectives expressed in natural language (e.g., "prioritise structural joints and load-bearing elements") and generates region-of-interest proposals annotated with semantic priority scores. These proposals modulate the information-gain objective of the Pred-NBV planner, biasing viewpoint selection toward semantically salient regions with high reconstruction uncertainty.

This closed-loop formulation transforms the exploration problem from a purely geometric optimisation into a *semantically informed* decision-making process. The VLA's spatial understanding—demonstrated at accuracies between 75.6% and 86.3% on spatial reasoning benchmarks [4]—enables the system to reason about occluded regions, predict the likely location of inspection-relevant features, and allocate observation budgets accordingly.

### B. Edge Computing Deployment

Deploying the proposed architecture on edge devices such as the NVIDIA Jetson Orin is essential for achieving true onboard autonomy in UAVs and mobile robots. The principal bottleneck is the memory footprint and inference latency of VLA models, which in their full-precision, cloud-deployed configurations require 40–80 GB of VRAM and exhibit inference latencies exceeding 2.50 seconds.

Two complementary techniques enable viable edge deployment. First, *Parameter-Efficient Fine-Tuning* via Low-Rank Adaptation (LoRA) reduces the number of trainable parameters by decomposing weight updates into low-rank matrices, enabling domain-specific adaptation (e.g., to infrastructure inspection) without retraining the full model. Second, *8-bit quantisation* compresses model weights from 32-bit floating point to 8-bit integers, reducing VRAM consumption by approximately $4\times$ with minimal accuracy degradation.

Table I summarises the expected performance characteristics under cloud and edge deployment scenarios.

TABLE I
Comparison of VLA Deployment Configurations

| Configuration | Method | Latency (s) | VRAM (GB) |
|---|---|---|---|
| Cloud (Traditional) | Full fine-tuning | $> 2.50$ | 40–80 |
| Edge (Optimised) | 8-bit + LoRA | 0.60–0.85 | 8–12 |

The 3DGS rendering pipeline itself is inherently suitable for edge deployment, achieving frame rates above 100 fps on modern GPUs [11]. Feed-forward reconstruction variants further reduce per-scene latency to sub-second levels (0.29–0.63 s), eliminating the iterative optimisation loop that would otherwise dominate on-board computation [11]. Zhang et al. [10] have demonstrated the feasibility of 3DGS-based semantic reconstruction on autonomous UAV platforms, validating that the computational requirements are compatible with current embedded hardware.

## IV. Comparative Analysis

This section presents a quantitative synthesis of performance metrics extracted from the surveyed literature, organised along four evaluation axes.

### A. Reconstruction Quality Metrics

The quality of 3DGS-based scene reconstruction is evaluated through three complementary metrics. *Peak Signal-to-Noise Ratio* (PSNR) measures pixel-level fidelity; in urban reconstruction tasks, 3DGS achieves values of up to 28.93 dB, surpassing classical multi-view stereo methods [12]. *Structural Similarity Index Measure* (SSIM) captures perceptual structural similarity, with 3DGS models attaining scores of 0.918 [3]. *Learned Perceptual Image Patch Similarity* (LPIPS) quantifies high-level perceptual differences; 3DGS achieves scores as low as 0.056, indicating photorealistic rendering quality that markedly exceeds NeRF baselines [3].

For geometric evaluation, the Absolute Trajectory Error (ATE) assesses localisation accuracy of the underlying SLAM system, while the Chamfer Distance (CD) measures the fidelity of the reconstructed point cloud against ground-truth geometry.

### B. Exploration Efficiency Metrics

Exploration efficiency is critical for energy-constrained UAV missions. The *Coverage Rate* and its integral, the Area Under the Coverage Curve (AUC), quantify how rapidly the planner achieves full object or scene coverage. Pred-NBV [1] demonstrates a 25.46% improvement in observed surface points relative to classical information-gain planners within identical time budgets.

The *Flight Path Length* versus coverage trade-off is equally important. GenNBV [7] optimises this ratio by learning a viewpoint selection policy that transfers across object categories, reducing total flight distance without sacrificing reconstruction completeness—a property directly translatable to extended UAV mission endurance.

### C. Cognitive Performance Metrics

The cognitive capabilities of VLA models are assessed through task-level success rates and spatial reasoning benchmarks. SwitchVLA [6] achieves task-switching success rates exceeding 90%, demonstrating robust multi-objective execution. On the EmbSpatial benchmark, Qwen2.5-VL [4] achieves spatial reasoning accuracies between 75.6% and 86.3%, while Lumo-1 [5] exhibits comparable performance with additional chain-of-thought reasoning capabilities.

### D. Hardware Feasibility Metrics

Hardware feasibility governs the practical deployability of the proposed architecture. 3DGS rendering achieves frame rates exceeding 100 fps on desktop GPUs and up to 200 fps in optimised configurations [11], comfortably above the 30 fps real-time threshold even when accounting for the reduced throughput of embedded GPUs. Feed-forward reconstruction methods achieve inference latencies of 0.29–0.63 seconds per scene [11], enabling quasi-real-time map updates.

As summarised in Table I, the combination of LoRA fine-tuning and 8-bit quantisation reduces VLA VRAM requirements from $\sim$80 GB to $\sim$8–12 GB, bringing them within the memory envelope of platforms such as the NVIDIA Jetson AGX Orin (32–64 GB unified memory).

## V. Discussion

## VI. Conclusion

This survey has reviewed the convergence of three transformative technologies—3D Gaussian Splatting for dense real-time reconstruction, Vision-Language-Action models for semantic cognition, and predictive Next-Best-View planning for efficient exploration—toward a unified framework for semantic-guided autonomous mapping.

The quantitative evidence consolidated from the literature demonstrates that 3DGS-based SLAM systems achieve photorealistic reconstruction quality (PSNR $\leq$ 28.93 dB, SSIM $\leq$ 0.918, LPIPS $\geq$ 0.056) at frame rates exceeding 100 fps, while VLA models attain spatial reasoning accuracies above 75% and task-switching success rates above 90%. Prediction-guided NBV planners further improve surface coverage by up to 25.46% under fixed time budgets.

Crucially, the combination of Low-Rank Adaptation and 8-bit quantisation reduces the computational footprint of VLA models to levels compatible with embedded platforms such as the NVIDIA Jetson Orin, with VRAM requirements of 8–12 GB and inference latencies below one second. This computational feasibility, coupled with the inherent efficiency of Gaussian-based rendering, establishes a credible pathway toward fully on-board, semantically aware autonomy for mobile robots and UAVs engaged in infrastructure inspection and autonomous exploration.

Future work should address the open challenges identified in Section V, with particular emphasis on safety-critical certification, multi-robot cooperative exploration, and the development of standardised benchmarks that jointly evaluate perception, cognition, and planning performance.

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